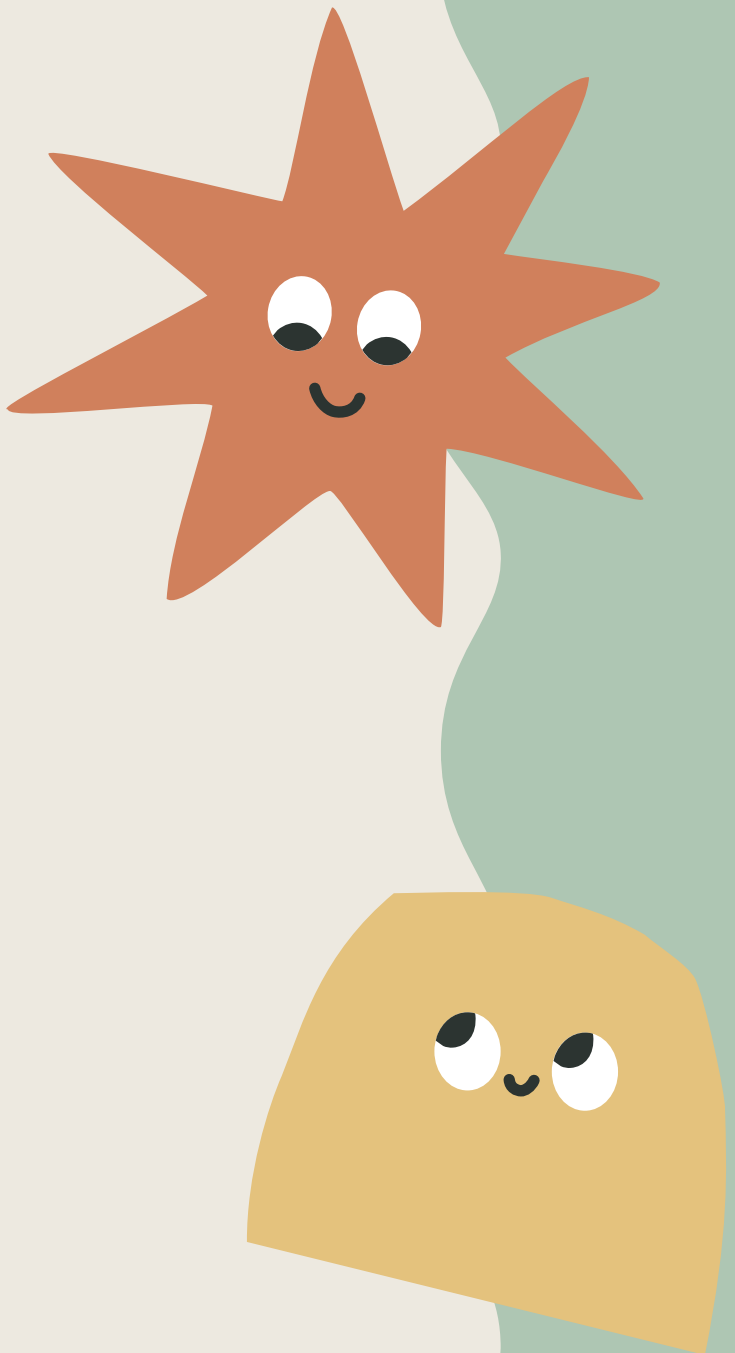
Propositional Logic in Artificial Intelligence

Lecture 6



What Is Logic?

- **Logic** is a formal system used to represent statements and determine whether conclusions follow from those statements
- Logic provides rules for reasoning correctly
- For example:
  - All humans are mortal
  - Ali is a human
  - Therefore, Ali is mortal
- Types of Logic
  - Propositional Logic
  - Predicate Logic

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Propositional Logic

- **Propositional Logic** is a formal system used to represent and reason about statements that can be either **True or False**
- It is also called **Propositional Calculus**
- Propositional logic represents knowledge using **propositions**
- So first we need to understand propositions

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What Is a Proposition?

- A **proposition** is a declarative statement that is either **True** or **False**
- Example:
  - Pakistan is in Asia
  - 2 + 2 = 5
  - The Earth revolves around the Sun
- Examples That Are Not Propositions
  - Close the door
  - Are you a student?
  - Is it raining?

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Proposition Symbols

- Instead of writing complete sentences repeatedly, we use symbols
- For example:
  - P = It is raining
  - Q = The road is wet
- Here:
  - P is a propositional variable
  - Q is another propositional variable
- The symbols themselves do not have a fixed meaning
- We can redefine meanings
  - P = Ali is a student
  - Q = Ali is intelligent

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Simple/Atomic vs Compound Propositions

- A single proposition that cannot be broken down into smaller propositions is called an **atomic proposition**
  - P = Ali is a student
  - Q = Lahore is in Pakistan
- **Compound propositions** are formed by combining atomic propositions using logical operators
  - P = It is raining
  - Q = The road is wet
  - We can say “It is raining AND the road is wet” symbolically
  - $P \wedge Q$
  - This is called a **compound proposition**

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Logical Connectives

➢Let P = "It is raining" and Q = "The ground is wet"

Symbol	Name	Meaning	Example
¬	NOT	Negation	$\neg P$ (It is not raining)
∧	AND	Conjunction	$P \wedge Q$ (It is raining AND the ground is wet)
∨	OR	Disjunction	$P \vee Q$ (It is raining OR the ground is wet)
→ OR ⟹	IMPLIES	Implication	$P \Longrightarrow Q$ (IF it is raining, THEN the ground is wet)
↔ OR ⟺	IFF	Biconditional	$P \Longleftrightarrow Q$ (It is raining IF AND ONLY IF the ground is wet)

NOT — Negation

- The symbol for NOT is: ¬
- Suppose: P = It is raining
- Then: ¬P means It is not raining
- NOT reverses the truth value.
- Truth Table

P	¬P
T	F
F	T

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AND — Conjunction

- The symbol for AND is: ∧
- Suppose: P = It is raining Q = The road is wet
- Then P ∧ Q means It is raining **AND** the road is wet
- AND is True **only when both propositions are True**
- Truth table

P	Q	P ∧ Q
T	T	T
T	F	F
F	T	F
F	F	F

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OR — Disjunction

- The symbol for OR is: v
- Suppose: P = It is raining Q = It is cloudy
- Then P v Q means It is raining **OR** it is cloudy
- OR is False **only when both sides are False**
- Truth table

P	Q	P v Q
T	T	T
T	F	T
F	T	T
F	F	F

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Implication

- The implication operator is: $\rightarrow$
- It is usually read as: if P, then Q
- Suppose: P = It is raining Q = The road is wet
- Then $P \rightarrow Q$ means If it is raining, then the road is wet
- The statement $P \rightarrow Q$ is False **only when P = True and Q = False**
- Truth table

P	Q	$P \rightarrow Q$
T	T	T
T	F	F
F	T	T
F	F	T

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Biconditional

- The biconditional operator is: ↔
- It is read as: If and only if
- P ↔ Q means P is true exactly when Q is true
- The statement is True **when both have the same truth value**
- Truth table

P	Q	P → Q
T	T	T
T	F	F
F	T	F
F	F	T

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Operator Precedence

- When multiple logical operators appear in one expression, we need to know which one is evaluated first
- A common order is:
  - Parentheses
  - NOT ¬
  - AND ∧
  - OR ∨
  - Implication →
  - Biconditional ↔

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Compound Logical Expression

- Suppose:
  - P = It is raining.
  - Q = The road is wet.
  - R = Traffic is heavy.
- We can create:
  - $(P \land Q) \rightarrow R$
- Read it as:
  - If it is raining AND the road is wet, then traffic is heavy
- This is a compound logical expression

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Evaluating a Logical Expression

- Suppose:
  - P = True
  - Q = True
  - R = False
- Expression: $(P \land Q) \rightarrow R$
  - First: $P \land Q$
  - Since both are True: $P \land Q$ = True
  - Now: True → False
  - Therefore: False
- So the entire expression is False

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Modus Ponens

- Modus ponens is a valid, basic rule of deduction in logic
- That means "if P implies Q, and P is true, then Q must also be true

$$\frac{P \qquad P \rightarrow Q}{Q}$$

Simple Example

- **Premise 1:** If it rains, the ground gets wet.
- **Premise 2:** It is raining.
- **Conclusion:** Therefore, the ground is wet.

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Advantages of Propositional Logic

- Simple and easy to understand using true/false statements
- Provides a clear and structured way to represent knowledge
- Logical expressions and truth tables can be processed by computer programs
- Useful for basic logical reasoning and decision-making
- Easy to implement in AI systems and rule-based models
- Acts as a foundation for more advanced logical systems

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Disadvantages of Propositional Logic

- It cannot naturally represent complex objects, properties, and relationships
- We cannot easily create a general rule such as:
  - All students are learners
- Becomes inefficient as truth tables grow exponentially with more propositions
- It does not have concepts such as **“all” (∀)** and **“some” (∃)**
- Works only with true/false values and cannot handle probabilities
- Cannot handle uncertainty or partial truth values

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